Yurou Celine Xiao

LinkedIn Google Scholar Ionic Stacks

SKILLS

Process & Hardware Design

- High-pressure high-temperature system design, pilot scaling, SolidWorks, engineering drawings, GD&T, laser cutter, drill press, soldering, electronics assembly

Electrochemistry & Fabrication

- LANDT Battery Test Systems, Metrohm Autolab, CC/CV, LSV, EIS, catalyst synthesis, electrode fabrication, electrolyte formulation

Analytical Chemistry & Software

- SEM, XRD, GC, IC, NMR, UV-vis, Python, MATLAB, OriginPro

EDUCATION

Ph.D. Mechanical Engineering – University of Toronto

- Electrochemical systems for integrated capture and conversion of CO_2

B.A.Sc. Mechanical Engineering – University of Alberta

Industry

MAY 2017 – AUG 2017

MODULAR CONSTRUCTION

Designed and modelled an industrial multi-function CNC machine to automate fastening and cutting processes.

SEPT 2017 – DEC 2017

SYNCRUDE CANADA LTD.

Engineered emergency and scheduled maintenance reliability packages for O&G and utility plants.

MAY 2018 – DEC 2018

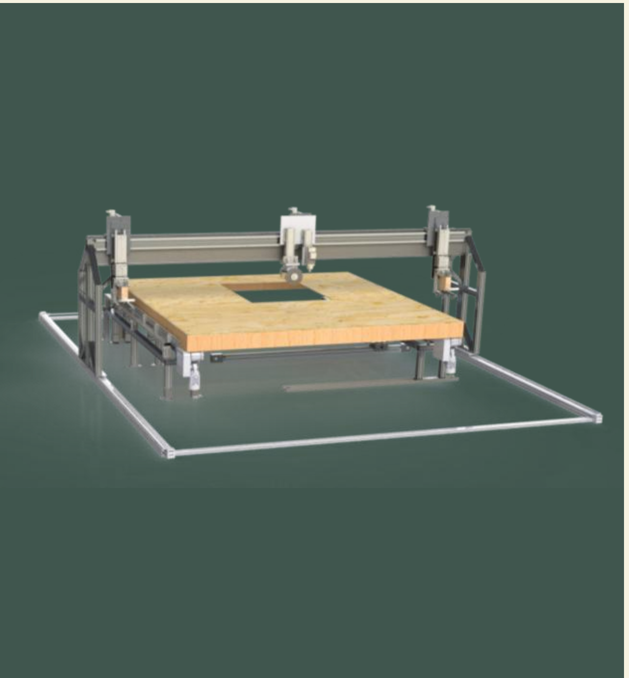
CITY OF EDMONTON

Acted as lead delegate in site meetings for municipal infrastructure projects.

AUG 2019 – DEC 2019

WSP CANADA

Transformed legacy P&IDs into a 3D-linked compliance tool and authored core engineering documentation.



Start-up

SEP 2020 – DEC 2020

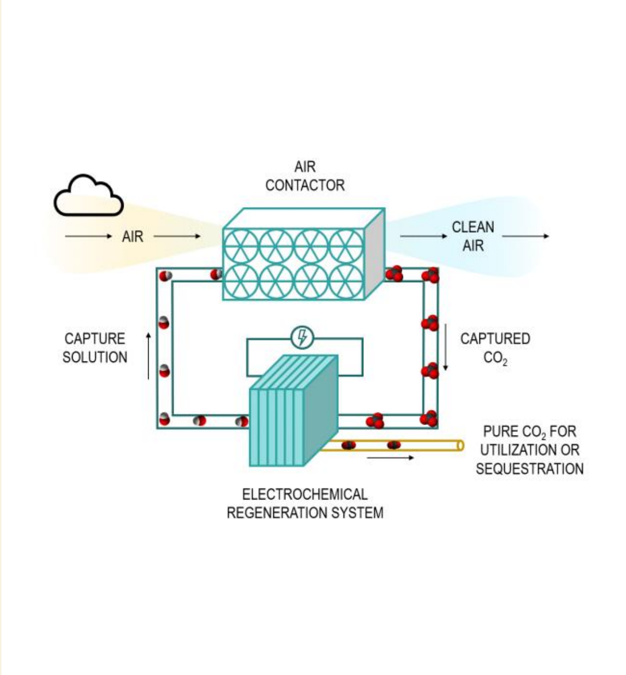
CERT SYSTEMS INC.

Scaled the electrochemical conversion of CO_2 into ethylene by operating a pilot plant (up to 40,000 cm^2 total active area), contributing to Team CERT's success as a Top 10 Finalist in the NRG COSIA Carbon XPRIZE.

SEP 2021 – APR 2022

E-QUESTER

Co-founded a startup team that developed a low-cost and scalable electrically powered direct air capture (DAC) system for CO_2 removal, resulting in a $250,000 Student Prize and named a Top 60 Milestone Award Finalist in the XPRIZE Carbon Removal competition.



Research

MAY 2020 – MAR 2021

UNIVERSITY OF CALGARY

Optimized energy performance for the net-zero MacKimmie Tower through thermal data analysis.

JAN 2024 – APR 2024

CSIRO ENERGY CENTER

Developed and scaled CO_2 electrolyzers from 1 cm^2 to 100 cm^2 for integrated CO_2 capture and conversion systems.

MAY 2023 – AUG 2025

SINTONLAB: PROGRAM LEAD

Directed a multi-year, $2M+ industry sponsored research program on integrated capture and conversion of CO_2.

SEP 2023 – AUG 2025

SINTONLAB: ELECTROLYZER DESIGNER

Managed electrolyzer Design for Manufacturing (DFM) as the technical liaison between R&D and machining teams.